

CAD / CAM CALIBRATION



MUSC JBECDM

Department of Reconstructive & Rehabilitation
Sciences

Division of Digital Dentistry

SYNOPSIS OF PROCEDURE

- CAD/CAM Restoration of a single anterior or posterior tooth utilizing e-max and either the Planmeca Fit system or the CEREC system.
- 3 hours to complete the procedure.
- Patient should pay prior to being brought back to the clinic for treatment.
- If significant decay exists in the tooth, such as might put the nerve at risk of an irreversible pulpitis, the decay should be removed and a build up performed at a procedure prior to the CAD/CAM visit. The final crown appointment should be delayed at least 3 weeks to see that the tooth remains asymptomatic.
- The expectation is that the preparation be complete 1:15 after the appointment begins.

ARMAMENTARIUM

- Digital scanner
- Depth cut bur
- 016 KR round ended shoulder diamond (medium and fine grit)
- Football diamond
- Critical thinking skills to determine best prep style (as taught in FP3)
- Isovac or dental dam for adhesive delivery of restoration
- Adhese Universal bonding agent/Variolink Esthetic resin cement

LITHIUM DISILICATE (E.MAX)

- Excellent track record over 10+ years.
- Good esthetics and available in 3 ranges of translucencies (HT, MT and LT).
- Allows for material thickness of 1mm when adhesively luted.
- Mills in 8-13 minutes.
- Crystallization cycle takes an additional 15 minutes.

Teichmann M. • Gockler F. • Weber V. • Yildirim M. • Wolfart S. • Edelhoff D.

Ten-year survival and complication rates of lithium-disilicate (Empress 2) tooth-supported crowns, implant-supported crowns, and fixed dental prostheses.

J Dent. 2017; 56: 65-77



DELIVERY OPTIONS NON-RETENTIVE PREPS

e.max

Luted

Bonded

tooth
clean and dry

restoration
X

tooth
clean and dry

restoration
20s

RMGI cement



Hydrofluoric acid etch



Bonding agent



20s



Silane

Tips

Dual-cured Resin Cement



1. Wave cure 1 second buccal and lingual

2. Keep continual seating pressure on restoration

3. Clean excess and floss after wave cure. Then cure 20s per surface

DELIVERY OPTIONS RETENTIVE PREPS

e.max/Empress

Panavia V5

tooth



20s

Primer

restoration



Hydrofluoric acid etch

20s e.max

60s Empress

60s Feldspathic



10s

Silane



Panavia V5

CLINICAL DECISION MAKING

- Evaluate tooth for proper material selection taking into account occlusion, location of finish lines, ability to adequately isolate for bonding.
- Evaluate tooth for preparation style. Is there an opportunity to be more conservative with the preparation or is a traditional crown the best treatment option?
- What is the best way to manage the tissue prior to taking an intraoral impression? Cord, laser, hemostatic agent...etc.
- What is the best way to isolate the area for delivery of the restoration and if you cannot isolate it, how would this change the procedure?

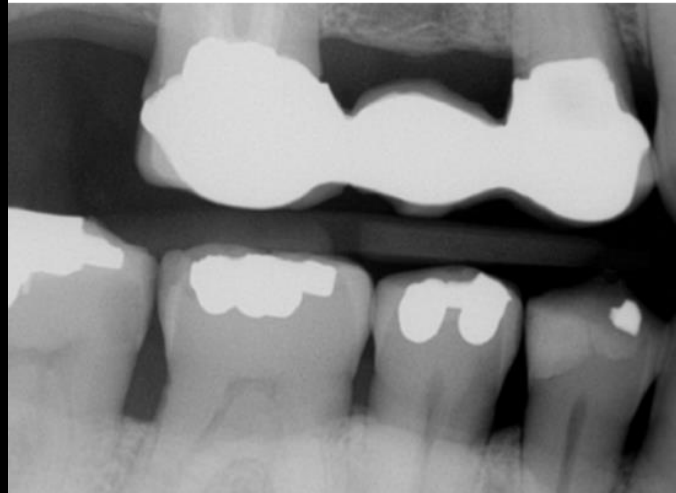
CLINICAL PROCEDURE

- Pay for the restoration upon bringing the patient back to the treatment area.
- Anesthesia
- Pick the shade of the restoration
- Check occlusion to see pre-operative occlusal situation.
- Preparation with a goal to complete in 60 minutes. Students are encouraged to keep finish lines supra-gingival and to be as conservative as possible within the limits of the restorative material they will be using.
- Tissue control
- IOS, design and mill.
- Try-in and adjust.
- Stain, glaze and crystallization of the e.max
- Isolation and delivery (generally accomplished by bonding the e.max)
- Check occlusion, contacts and assure no cement has been left behind
- Dismiss patient.

CONSIDERATIONS FOR CUSP FRACTURES

- Can an onlay be done?
- Anticipate thickness of remaining cusps once all old restorative material and decay is removed. If 2mm or greater, consider treatment planning for on onlay. If not, consider treatment planning for a crownlay. (see crownlay slide)

Case 1: Onlay



No buildup necessary

Temporize with Triage if necessary



CROWNLAY/OCCCLUSAL OVERLAY

A modern approach to a full coverage restoration. This preparation requires some fluidity in thinking. The goal is preservation of tooth structure, specifically the buccal and lingual axial walls. Keeping these high preserves the strength of the tooth, puts the margins on a thick band of enamel and keeps them in an area that is easily cleansable.

Step 1: mark occlusion and reduce the marked areas with depth cut bur.

Step 2: take out the old restoration and decay.

Step 3: break interproximal contact and prepare finish line on buccal and lingual height of contour.

Step 4: refine prep with fine diamonds

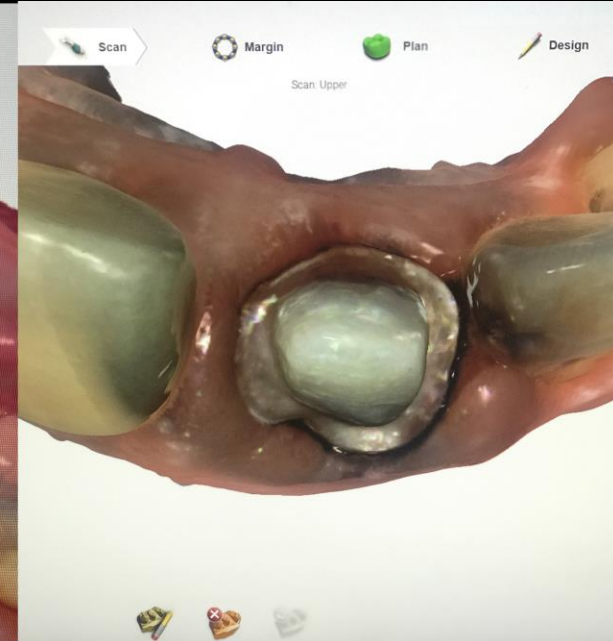


SCANNING TIPS

- Teeth must be kept extremely dry of saliva and blood during scanning. Both can obscure the details of the scan and make it unusable.
- Keep the scanner moving and watch the computer screen while scanning. Each scanner will give you an indication (audible tone for some, movement on the screen for others) of when data is being captured and when it is not.
- It is ok to stop the scan in the middle if the area is beginning to become wet again. Just dry and pick up where you left off.
- Refrain from having your loupe light or overhead light shining in the mouth during scanning.
- Pack 00 cord when finish lines are equi/sub gingival to help visualization of marginal area.



Heme obscuring
finish line

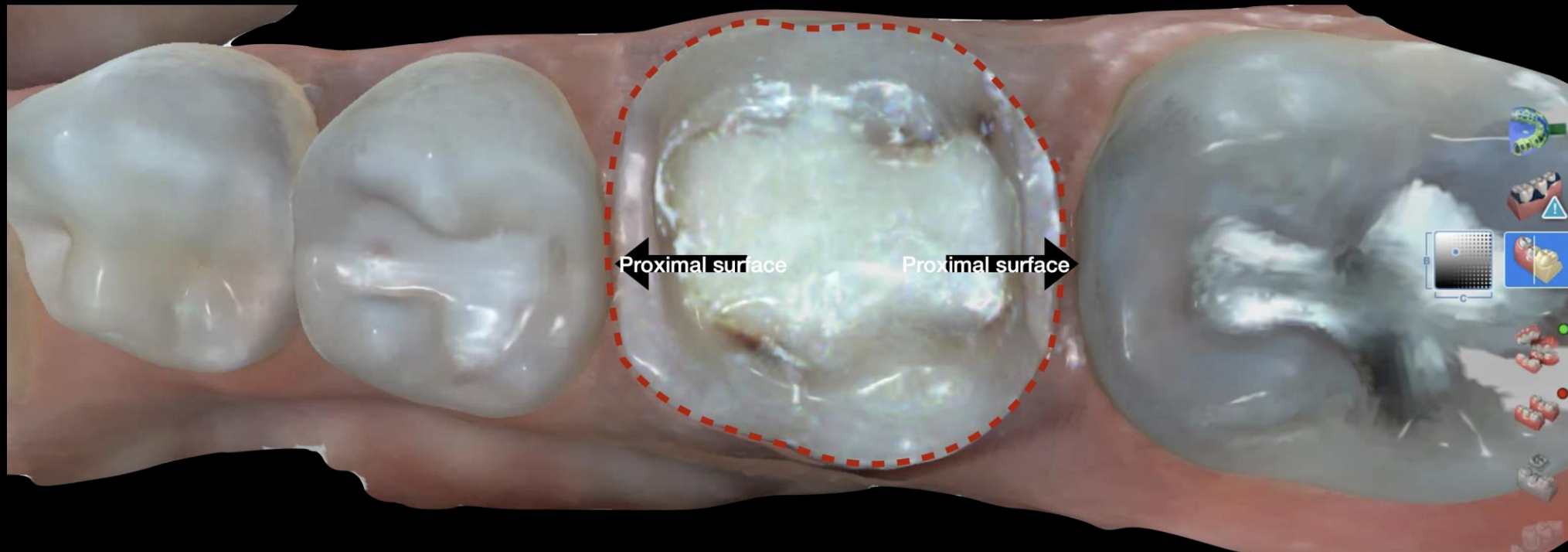


Good scan with
cord packed

INTRAORAL SCANNING

What is important in a digital impression?

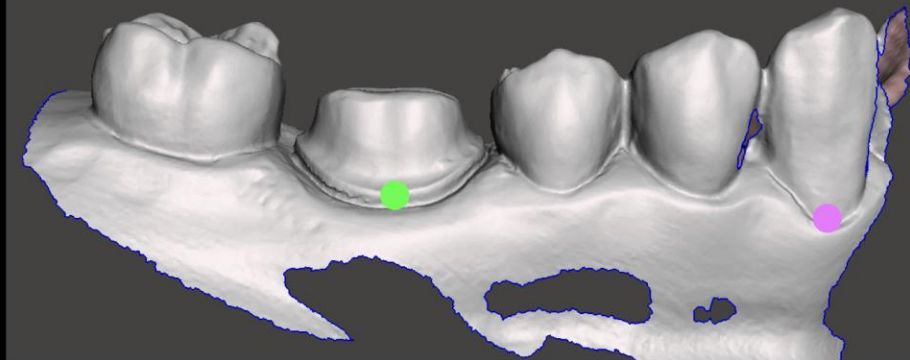
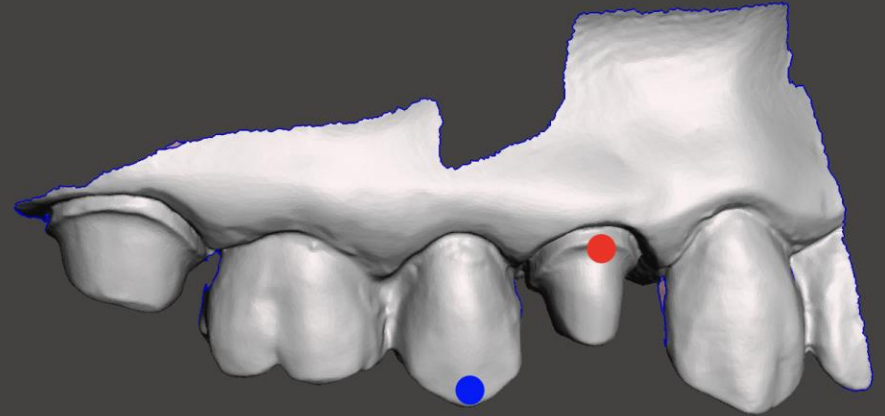
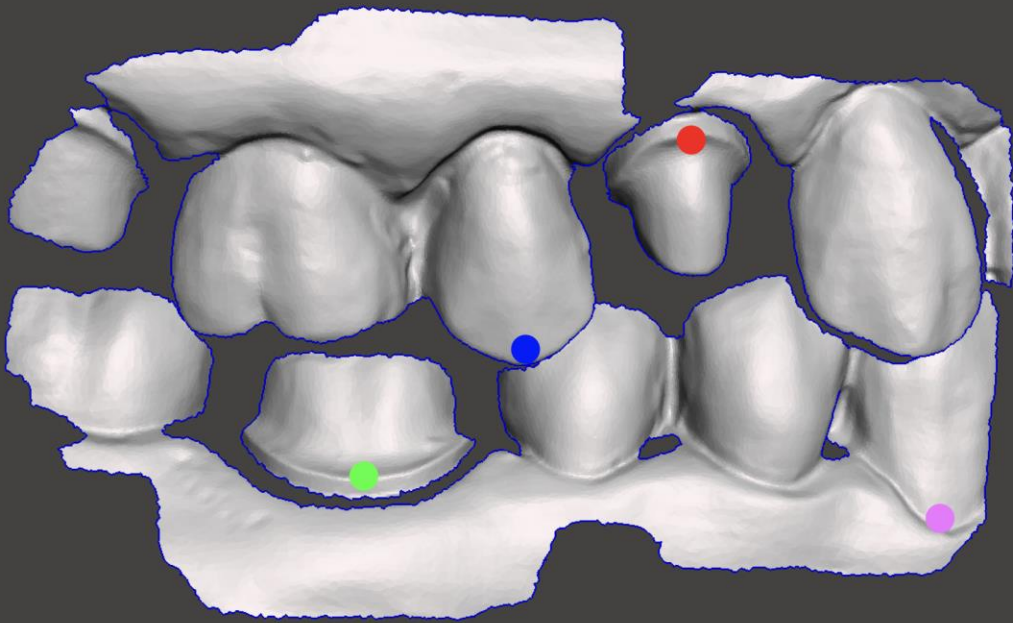
Span of a minimum of 4 teeth to aid in stitching the bite



Clean buccal surfaces on the prep and opposing scan

INTRAORAL SCANNING

Bite scan should include
teeth and tissue (2-3mm)



GRADING OF PROCEDURE

- Restorative Grading rubric used. Time management is a major factor in this clinic and stressed heavily.
- Major deductions for improper treatment planning. For example, if second molars are done in this clinic, prior consideration should be given to 1.ability to isolate for adhesive delivery 2.occlusion and amount of wear present in the dentition and 3.likely location of margins in relationship to the gingiva.
- Other common errors: Not breaking contact interproximally, finish lines on build-up material, insufficient moisture control during intraoral scanning and sharp transitional angles on the preparation.

SUPPLEMENTAL MATERIALS

Nonretentive Bonded Ceramic Partial Crowns: Concept and Simplified Protocol for Long-lasting Dental Restorations

Gianfranco Politano^a / Bart Van Meerbeek^b / Marleen Peumans^c

Purpose: Adhesively luted partial ceramic crowns have been documented to be clinically more durable than direct composite restorations when minimally invasively restoring large defects (replacing two cusps or more) in posterior teeth. The clinical longevity of such restorations is largely determined by the tooth-preparation design, material selection and adhesive luting procedure. The most frequently recorded failure in medium- to long-term clinical trials is fracture of the restoration. The clinical protocol of adhesively luted partial ceramic crowns can be optimized by taking the etiology of these restoration fractures into account. In this article, a simplified nonretentive bonded ceramic partial crown concept is presented that aims to achieve an adhesively luted ceramic restoration – composite cement – residual tooth structure biomechanical unit that maximally resists functional aging. Therefore, the three primary components of the bonded restoration-cement-tooth complex must function in synergy.

Methods, Results and Discussion: The clinical protocol starts with a tooth preparation designed to optimally absorb chewing stress. A stable, internally rounded and gently sloping tooth-preparation design with all outer margins inclined towards the tooth center assures a favorable and homogenous stress distribution with low cyclic fatigue subjected to the adhesive interface. This preparation form additionally enables the dental technician to fabricate a well-seating and -fitting ceramic restoration of uniform thickness. As restoration material, monolithic lithium-disilicate glass ceramic is sufficiently strong for the partial crown indication and preferred in order to decrease the fracture risk. Clinically essential for a long-lasting restoration is the optimal bond that can be obtained by combined micromechanical interlocking and chemical bonding of composite cement to hydrofluoric acid-etched and silanized glass ceramic.

Conclusion: The clinical effectiveness of this nonretentive bonded ceramic partial crown concept is confirmed by the overall high success rate as well as the very low fracture and debonding rate, as was recorded in long-term clinical trials.

Keywords: overlay, onlay, tooth preparation, glass ceramic, indirect restoration.

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Restoring posterior teeth exhibiting large defects (replacing two cusps or more) with directly bonded restorative composite is possible, but can be very challenging for

the dentist, in particular in busy clinical practices where treatment time should be as short as possible.⁶⁹ Upon a detailed biomechanical analysis of the remaining tooth structure, the tooth should be prepared minimally invasively, preserving as much sound tissue as possible, while removing any tooth parts with doubtful stability. In this sense, thin unsupported cusps must be reduced, as this will increase the durability of the restoration. The most difficult steps in the fabrication of large direct posterior composite restorations are the creation of a correct form with good occlusal anatomy as well as of well-contoured proximal surfaces with strong and correctly positioned contact points with the neighboring teeth. A posterior tooth restored in this direct composite workflow can function quite well in the medium term (ca 3–5 years).^{27,54,85} Bruxism and high caries risk seriously affect the durability of extensive composite restorations.^{68,92} Bonded indirect ceramic partial

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REVIEW ARTICLE

WILEY

Longevity of ceramic onlays: A systematic review

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Abstract

Objective: This systematic review aimed to evaluate the longevity of ceramic onlays and identify the factors that influence their survival.

Materials and methods: An electronic search was conducted through PubMed (MEDLINE), Google Scholar and Cochrane Library, up to August 2017. The literature search aimed to retrieve all the clinical studies on the longevity of ceramic onlays. Ceramic onlay was defined as any partial ceramic restoration that covers at least one cusp.

Results: A total of 21 studies met the selection criteria and were deemed suitable for this review. The medium-term studies (2–5 years) indicated a survival rate of 91–100%, and the long term studies (more than 5 years) showed a survival rate of 71–98.5%. The most common reason of failure was fracture, followed by debonding and caries. The most common patterns of deterioration were loss of margin integrity and discoloration. Onlay longevity can be enhanced if the preparation allows for at least 2 mm occlusal ceramic thickness and incorporates additional retentive features. Restoring teeth that are nonvital, teeth in a more posterior region, or teeth for patients with parafunctional habits appears to be associated with greater ceramic failure. Fabrication materials and methods, and adhesive bonding system did not seem to influence onlay longevity.

Conclusions: The clinical performance of the ceramic onlay appears acceptable regardless of the follow-up duration. Fracture of the ceramic onlay is the predominant cause of failure, and the most observed form of deterioration was associated with the restoration margin.

CLINICAL SIGNIFICANCE

Ceramic onlay appears to be a reliable option to restore posterior teeth. The most common pattern of failure is fracture of the ceramic material. The risk of ceramic onlay failure seems to increase if the restored tooth is nonvital and the patient demonstrates parafunctional habits.

1 | INTRODUCTION

Despite the outstanding performance of metal onlays and crowns,^{1–3} their unappealing appearance discourages patients from choosing them. Over the last 20 years, ceramic restorations have become very popular and routinely used in clinical practice. This is further driven by the significant developments that have improved the mechanical and optical properties of ceramic materials available for dental restorations.^{4–6} In addition, the development of modern manufacturing techniques has reduced the risk of internal flaw development within the ceramic material, which can further enhance its performance.^{7,8} In parallel to ceramic improvements, there have been advances in adhesive and cementation agents that combine enhanced bonding between

the tooth and the ceramic material and ease of use.^{9–11} Consequently, a new range of conservative, tooth-colored and durable restorative options are available.^{7,8,12,13} Contemporary ceramics have been used to restore teeth with inlays, onlays, crowns, or even fixed partial dentures.

The clinical studies have confirmed the successful use of ceramics as dental restorative materials.^{14,15} However, due to the considerable variation, it can be challenging to compare outcomes between studies, materials and restoration types. A number of reviews have collated this information to identify the success and survival of different ceramic restorations and also identify the biological and technical complications that exist.^{16–20} However, to the authors' knowledge, there are no reviews collating the clinical outcomes specifically of ceramic onlay

SUPPLEMENTAL MATERIALS

The compression dome concept: the restorative implications

Graeme Millich, BDS

Evidence now supports the concept that the enamel on a tooth acts like a compression dome, much like the dome of a cathedral. With an overlying enamel compression dome, the underlying dentin is protected from damaging tensile forces. Disruption of a compression system leads to significant shifts in load pathways. The clinical restorative implications are significant and far-reaching. Cutting the wrong areas of a tooth exposes the underlying dentin to tensile forces that exceed natural design parameters. These forces lead to crack propagation, causing flexural pain and eventual fracture and loss of tooth structure. Improved understanding of the microanatomy of tooth structure and where it is safe to cut teeth has led to a revolution in dentistry that is known by several names, including *microdentistry*, *minimally invasive dentistry*, *biomimetic dentistry*, and *bioemulation dentistry*. These treatment concepts have developed due to a coalescence of principles of tooth microanatomy, material science, adhesive dentistry, and reinforcing techniques that, when applied together, will allow dentists to repair a compromised compression dome so that it more closely replicates the structure of the healthy tooth.

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Key words: adhesive ceramic onlay, biodome, bioemulation, biomimetic, bio-rim, compression dome, minimally invasive

Anyone who has practiced dentistry for long enough recognizes that the conventional model of diagnosis and restoration proposed by G.V. Black does not serve well in the long term. When G.V. Black's restorative model was formulated, amalgam and gold were the primary restorative options, and the associated preparation techniques were based on the then available technologies and the requirements of the available restorative materials. The diagnostic model was the naked eye and a probe. Applying the current understanding of the ultrastructure of teeth in conjunction with modern diagnostic technologies, materials, adhesive techniques, and restorative approaches, dentists can now use minimally invasive biomimetic techniques to provide restorations that come close to functionally and biomechanically mimicking the original, healthy state of a tooth.

These advances started more than 25 years ago with a concept known as *microdentistry*, and one of the pioneers in this field was J.T. Rainey. His articles on dental microanatomy and micro-air abrasion were primary stepping stones and are now core to the biomimetic and bioemulation concepts currently taught in advanced restorative programs.^{1,2}

Concept of the compression dome

An early article describing the compression dome concept in association with biomechanical function addressed a previously unreported enamel fracture mechanism now described as *occlusal effect caries*.³ The problem is that teeth have been designed by nature primarily to withstand compressive loads. Teeth have not evolved to tolerate the damage that dentistry and high-speed



Fig 1. Stress patterns generated by a vertical load.¹ The direction of the lines indicates the direction of the stresses, obliquely vertical in the enamel compression dome and horizontal in the dentin core. Increased width of the fringes indicates increased stress. Stress concentration can be observed at the dentinoenamel complex. Stress concentration occurs where the fringes are closer together, a contributory factor to cervical abfraction.

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How the Variations of the Thickness in Ceramic Restorations of Lithium Disilicate and the Use of Different Photopolymerizers Influence the Degree of Conversion of the Resin Cements: A Systematic Review and Meta-Analysis

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Keywords

Ceramics; degree of conversion; lithium disilicate; resin cements.

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Abstract

Purpose: The purpose of this meta-analysis was to evaluate how the thickness variations in ceramic restorations of lithium disilicate and the use of different photopolymerizers influence the degree of conversion of the resin cements.

Methods: This study was performed according to the PRISMA checklist. The bibliographic research was performed to identify in vitro studies until December 2017 in the databases MEDLINE via PubMed, Web of Science, and Scopus. The selected studies were submitted to bias risk assessment. In the meta-analysis the data were evaluated using Review Manager Software v.5.3, with random-effect model at a 5% significance level.

Results: The meta-analysis in this study showed that despite the translucency of lithium disilicate, significant reduction of light was observed when activated through the ceramic. The degree of conversion was higher in the control groups in which the resin cement was photopolymerized without the interposition of the ceramic material.

Conclusions: The thinner the ceramic material, the higher the degree of conversion. A thickness greater than 1.0 mm drastically reduces the degree of conversion of dual-curing or light-curing resins. The curing unit with irradiance of 3200 to 3505 mW/cm² showed improvements in conversion values.